

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		Correct expression for centripetal force used (accept acceleration if equated to gravitational field strength) (1) accept with ω^2 Equating to gravitational expression i.e. $\frac{mv^2}{r} = \frac{GMm}{r^2}$ or $\frac{v^2}{r} = \frac{Gm}{r^2}$ (1) Enough algebra shown. Minimum is this: $\frac{mv^2}{r} = \frac{GMm}{r^2} \quad v^2 = \frac{GM}{r} \quad (1)$ Alternative 1: $m\omega^2 r$ or $\omega^2 r$ quoted (1) $m\omega^2 r = \frac{GMm}{r^2}$ and $v = \omega r$ or acceleration equivalent (1) Minimum algebra leading to answer (1) Alternative 2: Using M_1 or M_2 negligible in $T = 2\pi \sqrt{\frac{d^3}{G(M_1+M_2)}}$ (1) e.g. $2\pi \sqrt{\frac{d^3}{G(M_1+M_2)}} \approx 2\pi \sqrt{\frac{d^3}{GM}}$ Quoting $v = \omega r$ and $\omega = \frac{2\pi}{T}$ (1) or equivalent All algebra ok leading to the answer (1)	3			3	2	
	(b)		Measured velocities are too large (1) Accept velocity increasing on graph Link to the equation e.g. velocity increases with mass (1) Dark matter accounts for increased mass or the mass must be bigger (1)	3			3		